PRODUCT REQUIREMENTS DOCUMENT

ET AI Hackathon 2026

Problem Statement 6

AI for the Indian Investor

Opportunity Radar · Chart Pattern Intelligence · Market ChatGPT · AI Video Engine

Version	1.0 — March 2026
Hackathon	ET AI Hackathon 2026
Problem Statement	PS 6 — AI for the Indian Investor
Partners	Avataar.ai (Hiring Partner) · Unstop (Hackathon Partner)
Document Type	Full Product Requirements Document (PRD)
Scope	Opportunity Radar + Chart Pattern Intelligence (Primary), Market ChatGPT Next Gen (Secondary)
Target Users	Retail investors, F&O traders, mutual fund holders — India

1. Executive Summary

India has over 14 crore demat accounts, yet most retail investors operate without a structured intelligence layer. They react to WhatsApp forwards, miss critical regulatory filings, cannot interpret technical charts, and manage portfolios on instinct. Economic Times Markets sits on one of India's richest financial data assets. The gap is not data — it is actionable intelligence delivered at the right moment.

This product — working title InvestIQ — builds that intelligence layer: a multi-agent AI system that continuously monitors corporate signals, detects technical chart patterns, and delivers personalised, source-cited investment intelligence to retail and semi-professional investors across the ET ecosystem.

Core Value Proposition

From raw market data → to actionable trade intelligence in under 60 seconds.
14 crore+ demat holders get institutional-grade signal detection at zero marginal cost.
Back-tested pattern alerts, not guesswork. Every recommendation is source-cited and auditable.

2. Problem Statement

2.1 The Retail Investor Gap

Despite record demat account growth, the quality of retail investment decisions in India has not improved proportionally. Three structural problems persist:

- Information overload without signal extraction. Bulk deals, insider trades, SEBI filings, quarterly results, and management commentary are public — but buried in PDFs and BSE/NSE portals that retail investors cannot practically monitor.
- Technical analysis is a black box. Chart pattern recognition requires hours of screen time and expert knowledge most retail investors do not have. F&O traders especially suffer from late pattern identification.
- Generic advice, not portfolio-aware intelligence. Existing tools (news, apps, financial portals) deliver the same content to a student SIP investor and an active derivatives trader. Neither is well served.

2.2 Market Opportunity

Metric	Value	Source / Notes
Demat accounts (India)	14 crore+	SEBI, 2025
Active NSE traders (monthly)	~3.5 crore	NSE data
F&O participants	~60 lakh	NSE/BSE combined
ET Markets MAU	~4 crore	ET internal (approx.)
Retail investor avg. information lag	6–48 hours	vs. institutional

Addressable TAM (fintech signals)	$1.2B+	India fintech 2026

3. Product Vision & Goals

3.1 Vision Statement

To become the intelligence layer between Indian financial markets and the retail investor — making institutional-quality signal detection, pattern recognition, and portfolio-aware advice accessible to anyone with a demat account.

3.2 Success Metrics (Hackathon Demo Targets)

Metric	Baseline	Target (Demo)
Signal detection lag (filing → alert)	6–48 hours (manual)	< 60 seconds
Chart pattern back-test accuracy	N/A (manual)	> 70% historical hit rate
Relevant alerts per user per day	0 (generic news)	3–5 personalised
Multi-step analysis depth	Single query	5+ chained reasoning steps
Source citation rate	0%	100% of outputs

4. User Personas

Persona 1 — The Active F&O Trader

Attribute	Detail
Profile	28–42 years, metro city, trades Nifty/Bank Nifty options weekly
Tools used	Zerodha / Upstox, TradingView, WhatsApp groups
Pain point	Misses breakout entries; pattern spotted 2 hours too late
Goal	Real-time breakout + reversal alerts with back-tested confidence scores
Key feature	Chart Pattern Intelligence — breakout + divergence alerts on F&O stocks

Persona 2 — The Long-Term Equity Investor

Attribute	Detail
Profile	35–55 years, holds 8–15 stocks, checks portfolio monthly
Tools used	ET Markets, Moneycontrol, stockbroker app
Pain point	Misses insider trade filings and bulk deal signals until it's too late
Goal	Daily digest of signals relevant to stocks I hold, in plain language
Key feature	Opportunity Radar — filing alerts + management commentary shift

	detection

Persona 3 — The Mutual Fund / SIP Investor

Attribute	Detail
Profile	25–40 years, SIP in 3–5 funds, moderate financial literacy
Tools used	ET Money, Groww, ET Markets articles
Pain point	Cannot interpret market context; makes decisions based on gut or tips
Goal	Conversational Q&A about markets with portfolio-aware context
Key feature	Market ChatGPT Next Gen — portfolio-aware multi-step analysis

5. Feature Specifications

5.1 Opportunity Radar (Primary Feature)

A continuous monitoring agent that ingests structured financial signals from multiple sources and surfaces missed opportunities as prioritised, actionable daily alerts — not a summariser, but a signal-finder.

5.1.1 Data Sources Monitored

- BSE/NSE corporate filings — quarterly results, board meeting outcomes, insider trade disclosures
- Bulk and block deal feeds — real-time detection of large institutional moves
- Management commentary — earnings call transcript analysis for sentiment and guidance shifts
- SEBI regulatory actions — orders, penalties, approvals affecting listed companies
- Promoter shareholding changes — pledge creation/release, stake buy/sell
- FII/DII flow data — daily sectoral allocation shifts

5.1.2 Signal Classification Engine

Each detected event is classified across three dimensions before alerting:

Dimension	Classification Options	Example
Signal Type	Fundamental / Regulatory / Sentiment / Technical	Insider buy = Fundamental
Impact Direction	Bullish / Bearish / Neutral / Ambiguous	Promoter pledge = Bearish
Urgency	Immediate (< 1hr) / Same-day / Weekly watch	Bulk deal = Immediate
Portfolio Relevance	Held stock / Watchlist / Sector / Universe	User holds Infosys
Confidence Score	High / Medium / Low (based on historical signal reliability)	Insider buy = High

5.1.3 Alert Format

Sample Alert — Opportunity Radar Output

RELIANCE INDUSTRIES | Bullish | Confidence: High | Urgency: Same-day

Signal: Promoter entity acquired 12.3 lakh shares via bulk deal at Rs 2,847 on NSE (2.1x avg. daily volume).
Historical context: Last 4 similar promoter bulk buys on RIL preceded 8–14% gains over 90 days (3 of 4 instances).
Watch: Q4 results due in 11 days. Bulk buy ahead of results historically indicates confidence.
Source: NSE bulk deal data | ET Markets filing database

5.2 Chart Pattern Intelligence (Primary Feature)

Real-time technical pattern detection across the full NSE universe (2,000+ stocks + indices + F&O instruments), with plain-English explanation and historical back-tested success rates for that specific pattern on that specific stock.

5.2.1 Pattern Library (Phase 1 — Hackathon Scope)

Category	Patterns Detected
Breakout patterns	Cup & Handle, Flag, Pennant, Triangle breakout (ascending / descending / symmetrical)
Reversal patterns	Head & Shoulders (regular + inverse), Double Top / Bottom, Evening / Morning Star
Support / Resistance	Dynamic S/R levels, Historical pivot zones, VWAP deviation bands
Momentum indicators	RSI divergence (bullish / bearish), MACD crossover, Volume spike anomalies
Candlestick signals	Engulfing, Doji, Hammer, Shooting Star — confluence with trend direction

5.2.2 Back-Testing Methodology

- For each detected pattern, the system queries historical occurrences of the same pattern on the same ticker over the past 5 years.
- Calculates: hit rate (pattern led to expected move), average return, median time to target, and maximum drawdown during the pattern's resolution window.
- Outputs a confidence band: Strong (> 70% hit rate), Moderate (50–70%), Weak (< 50%) — Weak patterns are flagged but not alerted as primary signals.
- All back-test logic is auditable — users can drill into historical instances.

5.2.3 Pattern Alert Format

Sample Alert — Chart Pattern Intelligence

NIFTY BANK | Bullish Flag Breakout | Confidence: Strong (74% hit rate) | Timeframe: 1-hour chart

Pattern: Bullish flag formation resolved with volume confirmation. Breakout above 49,840.
Historical performance (last 5 years, same pattern on Bank Nifty): 74% led to 2.8–4.1% upward move in 3–7 sessions.
Measured target: 51,200 (3.5% upside). Stop-loss zone: 49,450 (flag base).
Note: FII bought Rs 2,140 Cr in financials today — directional confluence.
Disclaimer: This is pattern detection, not investment advice. Past performance is not indicative of future results.

5.3 Market ChatGPT — Next Gen (Secondary Feature)

A portfolio-aware conversational AI that handles multi-step market analysis queries, cites every data point, and reasons through investment scenarios like a structured analyst — not a news retriever.

5.3.1 Key Differentiators from Existing ET Market ChatGPT

Capability	Current ET Market ChatGPT	Next Gen (This Build)
Data depth	Single-source retrieval	Multi-source fusion (filings + prices + news)
Analysis steps	Single Q&A	5+ chained reasoning steps with intermediate outputs
Portfolio awareness	None	User portfolio context injected into every query
Source citation	Partial / none	100% — every claim linked to a source
Query handling	Simple factual	Comparative, scenario, what-if, back-test queries
Follow-up	Stateless	Conversation memory within session

5.3.2 Sample Supported Query Types

- Portfolio impact: 'If RBI cuts rates by 25bps next week, how does my portfolio get affected?'
- Comparative analysis: 'Compare HDFC Bank and ICICI Bank across capital adequacy, NIM, and NPA trend for last 3 quarters.'
- Signal synthesis: 'Are there any bullish signals on my watchlist stocks this week?'
- F&O specific: 'What is the options pain point for Nifty for this expiry based on OI data?'
- Back-testing: 'Has buying Infosys after earnings beats historically outperformed over 90 days?'

6. System Architecture

6.1 Agent Design

InvestIQ is built as a multi-agent system. Each agent has a defined role, tool set, and communication interface. Agents collaborate through an orchestration layer that manages task routing, parallel execution, and result synthesis.

Agent	Role	Tools / APIs
Ingestion Agent	Continuously pulls raw data from BSE, NSE, SEBI, ET Markets feeds	BSE API, NSE API, SEBI RSS, ET Markets internal
Signal Classifier Agent	Classifies each event by type, direction, urgency, confidence	LLM (GPT-4o / Claude), rule-based filters, sector taxonomy
Pattern Detection Agent	Runs technical analysis across OHLCV data streams	TA-Lib / custom Python, historical OHLCV store
Back-Test Agent	Queries historical pattern occurrence and computes performance stats	TimescaleDB / BigQuery, statistical compute
Portfolio Context Agent	Maintains user portfolio profile and injects context into all outputs	User demat data (CDSL/NSDL API or manual), watchlist DB
Alert Composition Agent	Generates human-readable, source-cited alerts in structured format	LLM with constrained output schema
Delivery Agent	Routes final alerts to correct channel (ET app, email, WhatsApp)	Push notification API, email, WhatsApp Business API
Conversation Agent	Handles multi-turn Market ChatGPT queries with session memory	LLM + RAG over ET Markets data, conversation store

6.2 Data Architecture

- Real-time stream: NSE/BSE bulk deal feeds, SEBI RSS, FII/DII data via scraping + official APIs — ingested every 60 seconds.
- Historical store: 5 years of OHLCV data for all NSE-listed stocks + indices, stored in a columnar time-series database (TimescaleDB or BigQuery).
- Filing corpus: Structured extraction of corporate filings (PDFs → structured JSON) using document AI pipeline.
- User profile store: Portfolio holdings, watchlist, risk preference, alert preferences — stored per user, encrypted.
- Pattern index: Pre-computed pattern scans updated every 15 minutes for all instruments, stored as a queryable pattern event log.

6.3 Technology Stack

Layer	Technology	Rationale
Orchestration	LangGraph / CrewAI	Multi-agent state management, retry logic

LLM backbone	GPT-4o / Claude Sonnet (via API)	Reasoning + structured output generation
Data ingestion	Python + Kafka streams	Real-time, fault-tolerant event processing
Technical analysis	TA-Lib + custom NumPy	Pattern detection, indicator computation
Time-series DB	TimescaleDB (PostgreSQL)	Efficient OHLCV queries + back-test scans
Vector store	Pinecone / Chroma	RAG over ET Markets content for ChatGPT agent
Backend API	FastAPI (Python)	Lightweight, async, OpenAPI-documented
Frontend	React + TailwindCSS	Demo dashboard for hackathon presentation
Delivery	Firebase (push) + SMTP + WA Business API	Multi-channel alert delivery

7. End-to-End Data Flow

7.1 Opportunity Radar Flow

NSE/BSE filing published → Ingestion Agent polls (60-second interval) → event parsed and normalised → Signal Classifier Agent runs LLM classification → Portfolio Context Agent checks user relevance → Alert Composition Agent drafts plain-English alert with source citation → Delivery Agent routes to user channel. Total latency target: < 90 seconds from filing publication.

7.2 Chart Pattern Flow

OHLCV stream updated (15-minute candles) → Pattern Detection Agent scans all instruments → matched patterns logged to pattern event store → Back-Test Agent queries historical performance for the same pattern × stock → confidence band computed → if Strong or Moderate, Alert Composition Agent generates alert → portfolio relevance filter applied → Delivery Agent routes. Additional full scans on daily open and close.

7.3 Conversational Query Flow

User submits query via chat interface → Conversation Agent extracts intent and entities → RAG retrieval over ET Markets corpus + live data fetch (if real-time data needed) → Portfolio Context Agent enriches query with user holdings → multi-step reasoning chain executed (up to 5 steps, each step output logged) → final answer composed with source citations → session stored for follow-up continuity.

8. Compliance & Risk Guardrails

Regulatory Compliance Framework

SEBI Investment Advisor regulations: InvestIQ is a signal and information tool, not a registered investment advisor. All outputs include mandatory disclaimer.

No personalized investment advice: System does not recommend specific buy/sell actions — it surfaces signals and patterns. Users make their own decisions.

Source citation mandatory: Every alert and analysis output must cite its data source. Unsourced claims are blocked at the composition layer.

Back-test disclaimer: All historical performance data is accompanied by standard past-performance disclaimer.

Data privacy: User portfolio data is encrypted at rest and in transit. Never used for model training without explicit consent.

Hallucination guardrail: LLM outputs are validated against retrieved source data before composition. Contradiction detection pipeline flags mismatches.

9. Impact Model

9.1 Quantified Value Estimates

Impact Dimension	Estimate	Assumptions
Information lag reduction	6–48 hrs → < 2 min	Automated filing ingestion vs. manual monitoring
Missed opportunity cost saved (per user/yr)	Rs 8,000–25,000	Avg. 2 missed bulk deal signals/month × avg. 4% move × Rs 50,000 position
Chart pattern alert value	Rs 3,000–12,000/yr	2 strong signals/month × 3% avg. return × Rs 50,000 deployed
Time saved on research	~4 hrs/week	vs. manual BSE/NSE portal monitoring
ET Markets engagement uplift	Est. 35–60% DAU increase	Personalised alerts drive daily return visits

9.2 Hackathon Demo Math

Back-of-envelope Impact Calculation

Target user segment: 50 lakh active ET Markets users with demat accounts
Addressable with signal alerts: 20 lakh (users trading individual stocks)
Average capital per user: Rs 2–5 lakh deployed
Expected signal quality improvement: 2–3 strong actionable signals/month vs. 0 currently
Conservative return improvement: 1.5% per annum from better entry/exit timing
Aggregate value created: 20L users × Rs 3.5L avg. capital × 1.5% = Rs 1,050 Crore / year
ET monetisation potential: Rs 999–2,999/year premium subscription × 5L paying users = Rs 500–1,500 Cr ARR

10. Build Roadmap

10.1 Hackathon Sprint Plan (48 Hours)

Phase	Hours	Deliverable
Phase 0: Setup	0–4 hrs	Repo, environment, API keys, data pipeline scaffolding
Phase 1: Data ingestion	4–10 hrs	NSE bulk deal feed + 1-year OHLCV data for 50 stocks ingested
Phase 2: Signal classifier	10–16 hrs	LLM-based filing classifier + alert composition working end-to-end
Phase 3: Pattern detection	16–28 hrs	3–5 patterns detected live on NSE universe with back-test stats
Phase 4: Chat interface	28–36 hrs	Market ChatGPT demo with 3 sample portfolio-aware queries
Phase 5: Demo polish	36–44 hrs	Frontend dashboard, alert UI, end-to-end demo flow
Phase 6: Submission	44–48 hrs	GitHub README, pitch video, architecture doc, impact model

10.2 Post-Hackathon Roadmap (If Productised)

- Month 1–2: Full NSE universe coverage, WhatsApp Business API integration for alerts, CDSL API integration for real-time portfolio sync.
- Month 3–4: AI Market Video Engine — auto-generate 30–90 second market update reels from alert data. FII/DII flow visualisations.
- Month 5–6: Premium tier launch (Rs 999/month) with advanced F&O signals — options pain point, OI-based support/resistance, PCR analysis.
- Month 7–12: Regional language support (Hindi, Tamil, Telugu) for alert delivery. Voice interface for market queries.

11. Open Questions & Dependencies

Question	Owner	Resolution Path
NSE/BSE API access for real-time data — cost and terms?	Tech lead	Use mock data for hackathon; explore NSE data vendor partnerships post-hackathon
SEBI compliance review of signal language?	Legal / Product	Add standard disclaimers; avoid 'buy/sell' language in all outputs
CDSL/NSDL API for portfolio sync — available?	Tech lead	Use manual portfolio input for hackathon demo; API integration in v2
LLM cost at scale (5L daily alerts)?	Product / Finance	Estimate: ~Rs 0.02/alert × 5L = Rs 10,000/day; offset by premium subscription revenue

WhatsApp Business API approval timeline?	Tech lead	Use email + push for hackathon; WhatsApp in post-hackathon sprint